

Grade 6 Accelerated
Unit 8
Lesson 1 Practice

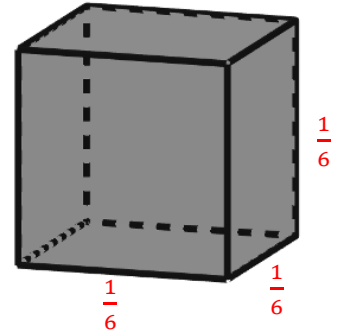
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A $\frac{1}{6}$ inch unit cube is shown.

1. Determine the volume of the unit cube. Include the units for the cube's volume.

$$V = lwh$$

$$V = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216} \text{ in}^3$$

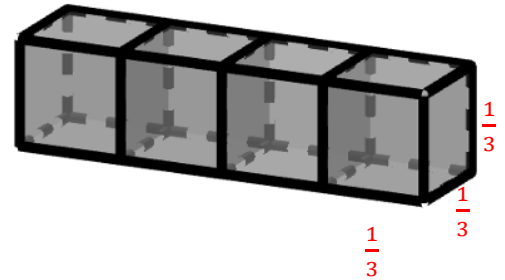


The diagram shows four unit cubes whose dimensions are $\frac{1}{3}$ inches.

2. The volume of one of the cubes is

$$\frac{1}{27} \text{ cubic inches (in}^3\text{)}.$$

$$\frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{27} \text{ in}^3$$



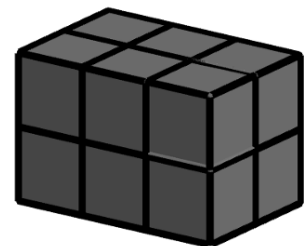
3. The volume of all four cubes is $\frac{4}{27} \text{ in}^3$.

$$4 \cdot \frac{1}{27} = \frac{4}{27}$$

A rectangular prism is shown.

4. How many unit cubes whose dimensions are $\frac{1}{2}$ inch fill up the rectangular prism?

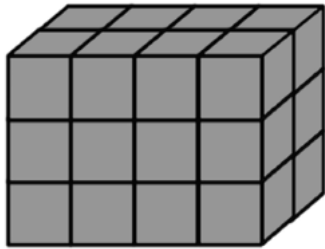
$$(2)(3)(2) = 12 \text{ cubes}$$



5. Each unit cube whose dimensions are $\frac{1}{2}$ inch has a volume of $\frac{1}{8}$ cubic inches. Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

$$12 \cdot \frac{1}{8} = \frac{12}{8} = 1\frac{1}{2} \text{ in}^3$$

6. Each unit cube in the diagram has dimensions of $\frac{1}{4}$ inch. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$.



$$\left(\frac{1}{4}\right)^3 = \frac{1}{64} \text{ in}^3 \text{ per cube}$$

$$(4)(2)(3) = 24 \text{ cubes}$$

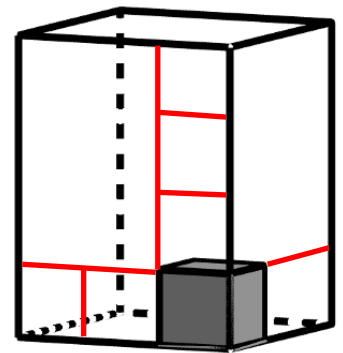
$$V = \frac{1}{64} \cdot 24 = \frac{24}{64} = \frac{3}{8} \text{ in}^3$$

A rectangular prism with one cube inside is shown.

7. How many cubes will it take to fill the rectangular prism?

$$(3)(2)(4) = 24 \text{ cubes}$$

8. The rectangular prism has a length of $\frac{9}{4}$ centimeters (cm), a width of $\frac{6}{4}$ cm, and a height of 3 cm as shown. Determine the dimensions of the fractional unit cube.



Each unit cube has dimensions of $\frac{3}{4}$ in.

9. Determine the volume of the unit cube.

$$\left(\frac{3}{4}\right)^3 = \frac{27}{64} \text{ in}^3$$

10. Determine the volume of the rectangular prism.

$$V = \frac{27}{64} \cdot 24 = \frac{648}{64} = 10.125 \text{ in}^3$$

OR

$$V = \left(\frac{9}{4}\right)\left(\frac{6}{4}\right)(3) = 10.125 \text{ in}^3$$

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A rectangular prism is shown with units measured in inches (in.).

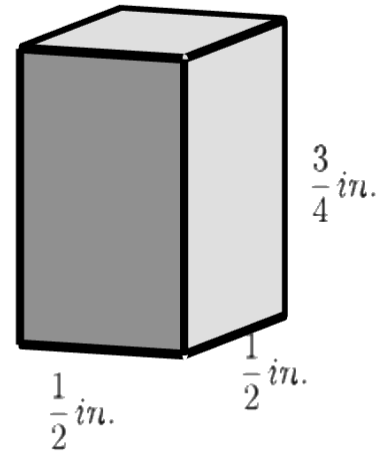
In questions 1-4, find the volume using cubes with side lengths $\frac{1}{4}$ in.

1. Determine how many cubes will fit along the length, width, and height.

$$\frac{1}{2} \div \frac{1}{4} = \underline{2} \text{ cubes will fit along the length.}$$

$$\frac{1}{2} \div \frac{1}{4} = \underline{2} \text{ cubes will fit along the width.}$$

$$\frac{3}{4} \div \frac{1}{4} = \underline{3} \text{ cubes will fit along the height.}$$



2. A total of 12 cubes will fit within the rectangular prism.

3. The volume of each cube is $\frac{1}{64} \text{ in}^3$. Multiply the number of cubes needed to fill the prism by the volume of each cube to find the volume of the prism. Include units.

$$12 \left(\frac{1}{64} \right) = \frac{12}{64} = \frac{3}{16} \text{ in.}^3$$

4. Instead of using cubes, Kaia uses the formula $V = l \times w \times h$. Fill in the spaces below to complete her work. Include units.

$$V = \underline{\frac{1}{2}} \times \underline{\frac{1}{2}} \times \underline{\frac{3}{4}}$$

$$V = \underline{\frac{3}{16}} \text{ in.}^3$$

A miniature Rubik's cube has side lengths 0.8 inches (in.). The miniature cubes can be packed into a larger box with dimensions of 15 in. $\times$ 15 in. $\times$ 10 in.

5. How many miniature cubes can be packed into the larger box?

$$15 \div 0.8 = 18.75$$

$$18.75 \times 12.5 \times 18.75 = 4,394.53125$$

$$10 \div 0.8 = 12.5$$

$$15 \div 0.8 = 18.75$$

6. Use the number of miniature cubes that can be packed in the larger box to determine the volume of the box. Include units.

$$4,394.53125(0.8^3) = 2,250 \text{ in}^3$$

Chera and Dylan are calculating the volume of a rectangular prism with the dimensions of 1.5 centimeters (cm) $\times$ 1.5 cm $\times$ 3 cm using different size cubes. Chera uses cubes with side lengths lengths $\frac{1}{4}$ cm, and Owen uses cubes with side lengths $\frac{1}{2}$ cm.

7. Who will need more cubes to fill the larger prism? Justify.

Chera, because her cubes have a smaller volume.

$$\left(\frac{1}{4}\right)^3 = \frac{1}{64} \text{ cm}^3 < \left(\frac{1}{2}\right)^3 = \frac{1}{8} \text{ cm}^3$$

8. What is the volume of the rectangular prism? Include units.

$$1.5 \times 1.5 \times 3 = 6.75 \text{ cm}^3$$

9. If they had used a cube with side lengths $\frac{3}{4}$ cm instead, would they have found the same volume of the rectangular prism?

Yes

10. A sandbox in the shape of a rectangular prism is being filled with sand. The dimensions of the sandbox are 32.25 inches (in.) by 44.5 in. If the height of the sand is 15 in., what is the volume of the sand in cubic inches?

$$32.25 \times 44.5 \times 15 = 21,526.875 \text{ in}^3$$

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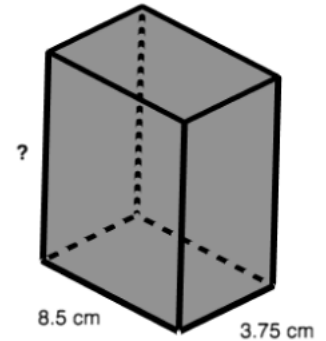
A rectangular prism is shown with measurements in centimeters (cm).

1. What is the area of the bottom face of this prism? Include units.

$$8.5 \times 3.75 = 31.875 \text{ cm}^2$$

2. If the volume of the box is 382.5 cubic centimeters (cm^3), what is the value of the missing dimension of the box?

$$382.5 = 31.875 \cdot h \Rightarrow h = 12 \text{ cm}.$$



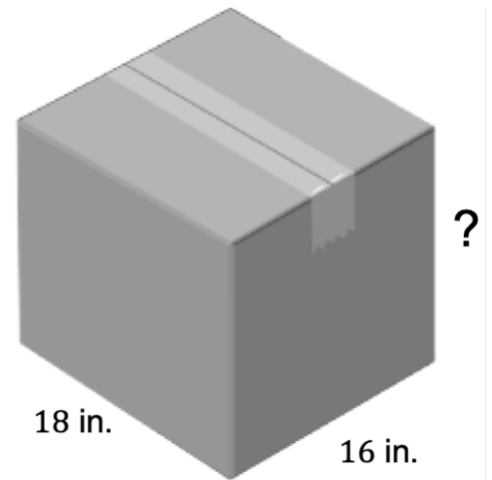
Sarai is using moving boxes from the post office. The boxes have a length of 18 inches (in.) and a width of 16 in., but she cannot recall the height of the box

3. Determine the area of the bottom face of the box. Include units.

$$18 \times 16 = 288 \text{ in}^2$$

4. If the volume of the box is 5,184 cubic inches (in^3), what is the height of the box? Include units.

$$5184 = 288h \Rightarrow h = 18 \text{ in}$$



Jenni was trying to find the missing dimension of the lunchbox shown. She made a mistake when solving for the width of the box, but she cannot find it. Her work is shown.

Volume = 234 cubic in.



Jenni's work	
$\begin{array}{r} 6.5 \\ \times \quad 9 \\ \hline 585 \end{array}$	$234 \div 5.85 = 40$ $w = 40 \text{ in.}$
$585 \rightarrow 5.85 \text{ in}^2$	

5. Circle the mistake.

6. What should the width of the box be? Include units.

$$234 \div 58.5 = 4 \Rightarrow w = 4 \text{ in.}$$

7. Write a note to Jenni on how to correct her error(s).

The decimal point should be moved one decimal place to the left.

$$585 \Rightarrow 58.5$$

8. A pizza box has a length of 14 inches (in.) and a height of 2.25 in. The box has a volume of 441 cubic inches (in^3). What is the width of the box? Include units.

$$441 = (14 \cdot 2 \cdot 25) \cdot w$$

$$441 = 31.5 \cdot w \Rightarrow w = 14 \text{ in.}$$

9. Mike bought an aquarium. He knows that the length is 36 inches (in) and the width is 18.25 in., but he can't remember the appropriate height. The volume needs to be 11,169 cubic inches (in^3). What is the height of the aquarium? Include units.

$$11,169 = (36 \cdot 18.25) \cdot h$$

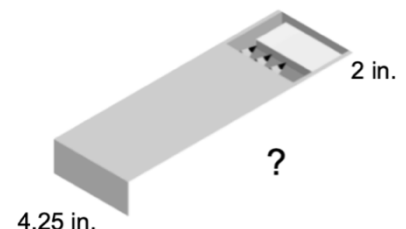
$$11,169 = 657h \Rightarrow h = 17 \text{ in.}$$

10. Sharee is measuring her pencil case. The case has a width of 4.25 inches (in.) and a height of 2 in. The volume of the case is 68 cubic inches (in^3). Determine the length of the suitcase. Include units.

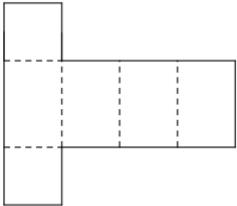
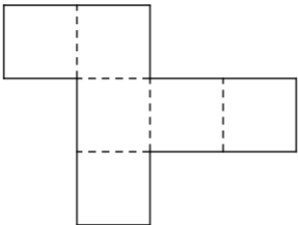
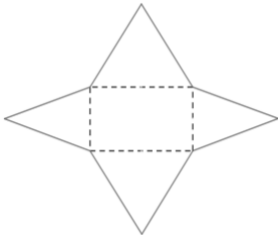
$$68 = (4.25 \cdot 2) \cdot \ell$$

$$68 = 8.5 \cdot \ell$$

$$8 = \ell \Rightarrow 8 \text{ in.}$$

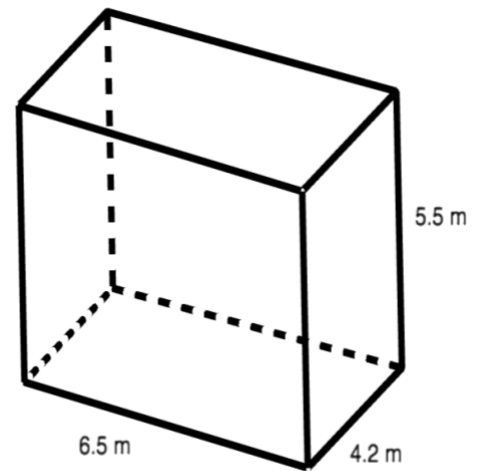
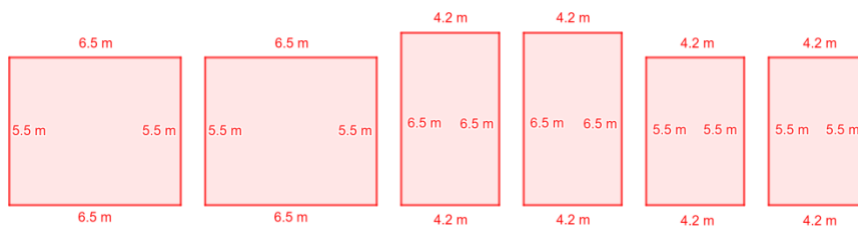


Complete the table to describe the three-dimensional figure formed for questions 1-4.

	Net	Three-Dimensional Figure Formed
1.		Rectangular Prism
2.		Pyramid
3.		Cube
4.		Pyramid Rectangular Pyramid

A three-dimensional figure is shown for questions 5-6.

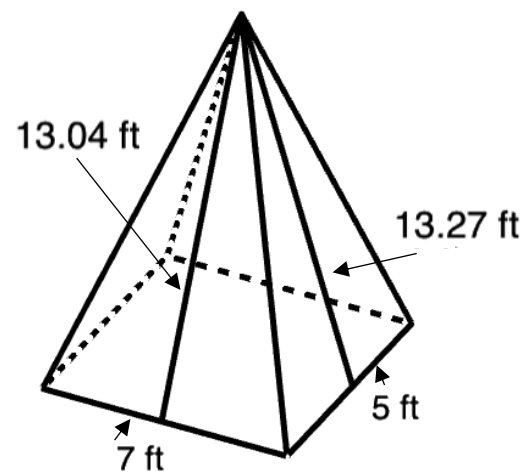
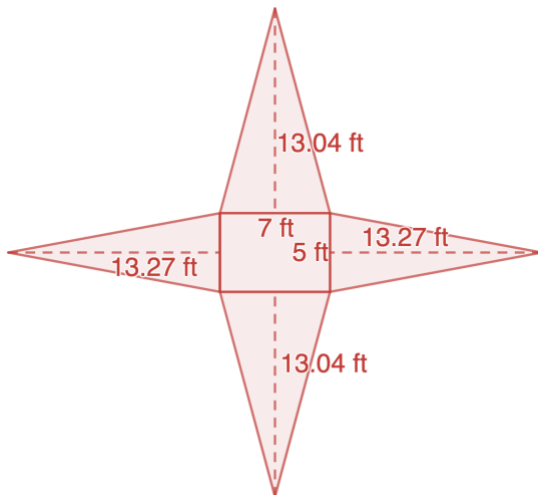
5. Sketch the net or the individual faces of the three-dimensional figure.



6. Label each side of the net with the appropriate measurement in meters (m).

A three-dimensional figure is shown for questions 7-8.

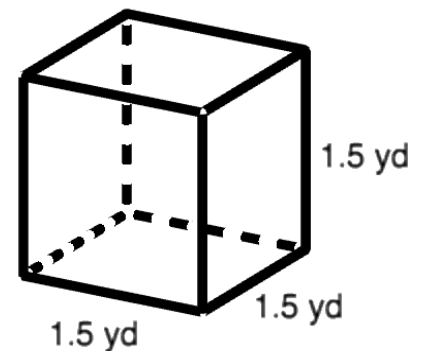
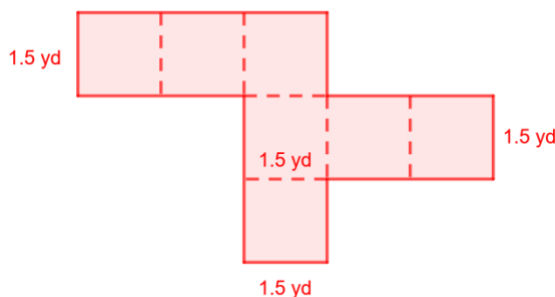
7. Sketch the net or the individual faces of the three-dimensional figure.



8. Label each side of the net with the appropriate measurement in feet (ft).

A three-dimensional figure is shown for questions 9-10.

9. Sketch the net or the individual faces of the three-dimensional figure. ***all dimensions are 1.5 yd**



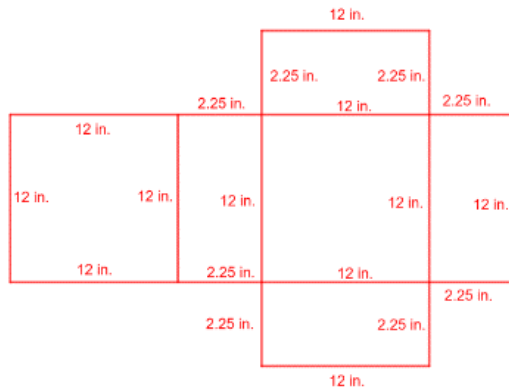
10. Label each side of the net with the appropriate measurement in yards (yd).

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A pizza box measures 12 inches (in.) long, 12 in. wide, and 2.25 in. tall.

1. Sketch the net of the pizza box.



2. Label the dimensions of each face on the net.

3. Find the area of each face. Include units.

1. $12 \times 12 = 144 \text{ in.}^2$
2. $12 \times 12 = 144 \text{ in.}^2$
3. $12 \times 2.25 = 27 \text{ in.}^2$

4. $12 \times 2.25 = 27 \text{ in.}^2$
5. $12 \times 2.25 = 27 \text{ in.}^2$
6. $12 \times 2.25 = 27 \text{ in.}^2$

4. Add the area of the faces to find the surface area of the pizza box. Include units.

$$144 + 144 + 27 + 27 + 27 + 27 = 396 \text{ in.}^2$$

5. Geri is wrapping a small gift box for a friend's birthday. The box is 2.75 inches (in.) long, 1.75 in. wide, and 1 in. high. What is the surface area of the box that Geri will need to buy wrapping paper for? Include units.

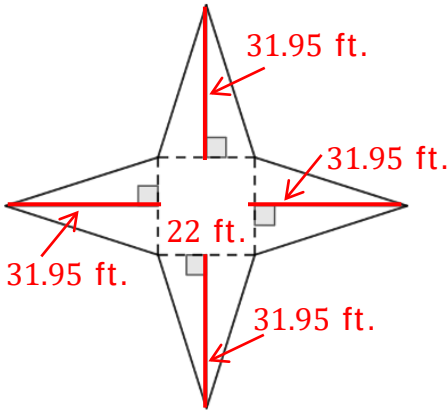
1. $2.75 \times 1.75 = 4.8125 \text{ in.}^2$
2. $2.75 \times 1.75 = 4.8125 \text{ in.}^2$
3. $2.75 \times 1 = 2.75 \text{ in.}^2$
4. $2.75 \times 1 = 2.75 \text{ in.}^2$
5. $1.75 \times 1 = 1.75 \text{ in.}^2$
6. $1.75 \times 1 = 1.75 \text{ in.}^2$

$$4.8125 + 4.8125 + 2.75 + 2.75 + 1.75 + 1.75 = 18.625 \text{ in.}^2$$

In the area that is now known as the Sudan, there is a collection of pyramids known as the Nubian Pyramids. The largest pyramid has a square base of 22 feet (ft) and stands 30 ft high. The slant height is 31.95 ft. The net of the pyramid is shown.

6. Label the dimensions of the net. Include units.
7. Complete the table to determine the area of each face of the pyramid.

Side	Formula	Area
Rectangular Base	$A = l \cdot w$	$A = 22 \times 22 = 484 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$



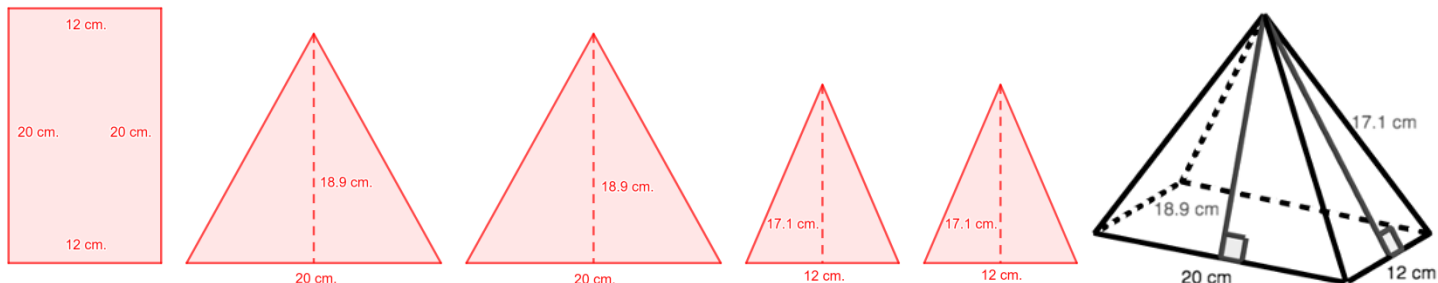
8. Determine the surface area of the pyramid. Include units.
 $484 + 351.45 + 351.45 + 351.45 + 351.45 = 1,889.8 \text{ ft.}^2$
9. Calculate the surface area of a rectangular pyramid that has a square base with a side length of 11.5 inches (in.), which is the base of the triangle with a slant height of 15.13 in.
 $11.5 \times 11.5 = 132.25 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
 $\frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2$
 $132.25 + 4(86.9975) = 480.24 \text{ in.}^2$
10. Calculate the surface area of a rectangular pyramid that has a rectangular base with a side length of 4 feet (ft), which is the base of a triangle with a slant height of 3.9 ft, and a side length of 5 ft, which is the base of a triangle with a slant height of 4.1 ft.
 $4 \times 5 = 20 \text{ ft.}^2$
 $\frac{1}{2} \times 4 \times 3.9 = 7.8 \text{ ft.}^2$
 $\frac{1}{2} \times 4 \times 3.9 = 7.8 \text{ ft.}^2$
 $\frac{1}{2} \times 5 \times 4.1 = 10.25 \text{ ft.}^2$
 $\frac{1}{2} \times 5 \times 4.1 = 10.25 \text{ ft.}^2$
 $20 + 7.8 + 7.8 + 10.25 + 10.25 = 56.1 \text{ ft.}^2$

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A rectangular pyramid is shown with measurements in centimeters (cm).

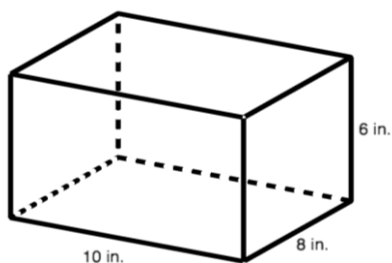
1. Draw a net or the individual faces of the figure. Label the measurements.



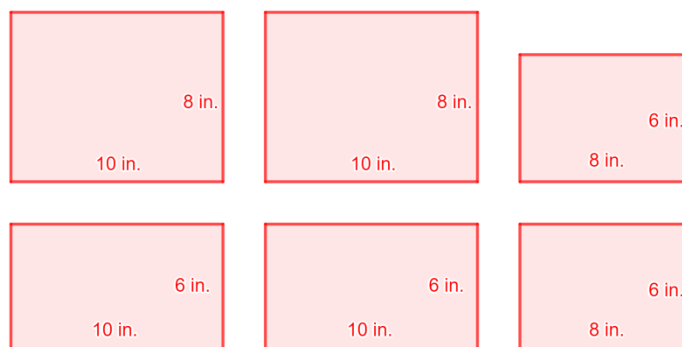
2. Find the surface area of the figure.

$$\begin{aligned}
 1. \quad & 12 \times 20 = 240 \text{ cm.}^2 & 4. \quad & \frac{1}{2}(12)(17.1) = 102.6 \text{ cm.}^2 \\
 2. \quad & \frac{1}{2}(20)(18.9) = 189 \text{ cm.}^2 & 5. \quad & \frac{1}{2}(12)(17.1) = 102.6 \text{ cm.}^2 \\
 3. \quad & \frac{1}{2}(20)(18.9) = 189 \text{ cm.}^2 & & 240 + 2(189) + 2(102.6) = 823.2 \text{ cm.}^2
 \end{aligned}$$

A rectangular prism is shown with measurements in inches (in.).



3. Draw a net or the individual faces of the figure. Label the measurements.



4. Find the surface area of the figure.

$$\begin{aligned}
 & (10 \times 8) + (10 \times 8) + (8 \times 6) + (8 \times 6) + (10 \times 6) + (10 \times 6) \\
 & = 80 + 80 + 48 + 48 + 60 + 60 = 376 \text{ in.}^2
 \end{aligned}$$

5. Find the volume of the figure.

$$\begin{aligned}
 V &= \ell \times w \times h \\
 V &= 10 \times 8 \times 6 \\
 V &= 480 \text{ in.}^3
 \end{aligned}$$

Tony designs earrings and is making post earrings in the shape of rectangular pyramids. They have a square base of 4 millimeters (mm). The slant height is 3.6 mm.

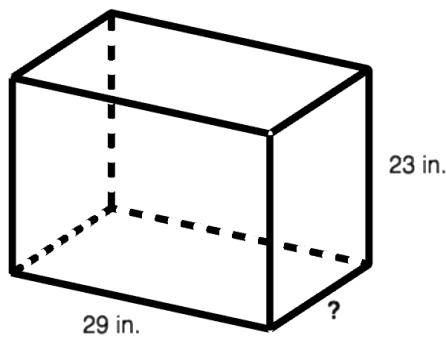
6. What is the surface area of one earring?

$$\begin{aligned} & (4 \times 4) + \left(\frac{1}{2} \times 4 \times 3.6\right) + \left(\frac{1}{2} \times 4 \times 3.6\right) + \left(\frac{1}{2} \times 4 \times 3.6\right) + \left(\frac{1}{2} \times 4 \times 3.6\right) \\ &= 16 + 7.2 + 7.2 + 7.2 + 7.2 \\ &= 44.8 \text{ mm.}^2 \end{aligned}$$

7. What is the surface area of a pair of earrings?

$$2(44.8) = 89.6 \text{ mm.}^2$$

Delilah is designing a toy box for her cousin. She drew a model of the toybox.



8. If the case has a volume of 12,673 cubic inches (in^3), what is the width of the box?

$$\begin{aligned} V &= \ell \times w \times h & 12,673 &= 667w \\ V &= 29 \times w \times 23 & \Rightarrow w &= 19 \text{ in.} \\ V &= 667w \end{aligned}$$

9. How much wood would Delilah need to make a box with these dimensions, not taking into consideration any overlap of the faces?

$$\begin{aligned} & (29 \times 19) + (29 \times 19) + (19 \times 23) + (19 \times 23) + (29 \times 3) + (29 \times 3) \\ &= 551 + 551 + 437 + 437 + 667 + 667 \\ &= 3310 \text{ in.}^2 \end{aligned}$$

10. If the inside of the box has a length of 28 in., a width of 16.5 in., and a height of 17 in., how much smaller is the volume of the inside of the toybox than the volume of the entire toybox?

$$\begin{aligned} V_{\text{inside}} &= 28 \times 16.5 \times 17 = 7,854 \text{ in.}^3 \\ 12,673 - 7,854 &= 4,819 \text{ in.}^3 \end{aligned}$$